

Determining the molecular diffusion coefficient of naphthenic acid in water as a heavy oil pollutant by experimental method

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ABSTRACT

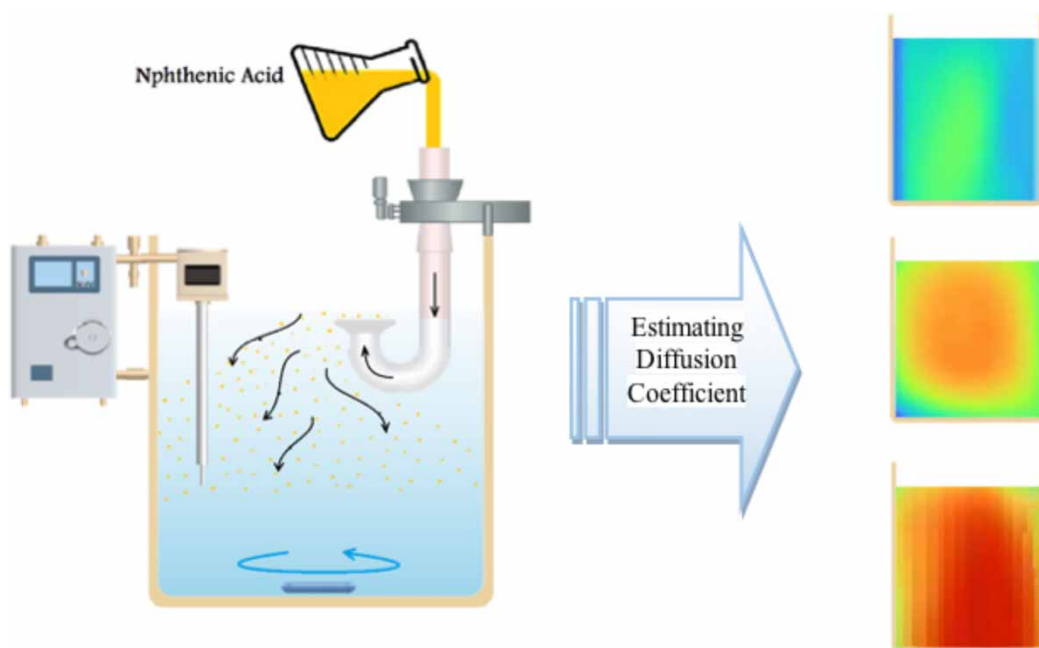
Naphthenic acids are the most significant environmental pollutants created by the extraction of oil from oil sand deposits. Since the diffusion coefficient of naphthenic acid has a direct effect on the modeling of diffusion and advection and determining its behavior and movement in water, this number is needed for modeling work and future studies. In this study, the diffusion coefficient of this substance in water was determined experimentally and in a laboratory. The researchers used the device of the Armfield company, known as the device for determining the diffusion coefficient in liquids. In this research, after examining different methods of determining the concentration of naphthenic acid in the water, including UV-vis, chemical oxygen demand (COD), crystal violet, oxidation reduction potential, pH meter, and electrical conductivity meter, the COD method was the best method in determining the concentration at different times, which provided a suitable numerical range for the concentrations and a device was built for heavy oil pollutants with poor solubility in water to specify the diffusion coefficient, and for the first time, the diffusion coefficient of naphthenic acid in water was obtained with experiments and experimental equations as $0.69 \times 10^{-9} \text{ m}^2/\text{s}$ which indicates the weak diffusion of this substance in water.

Key words: chemical oxygen demand, diffusion coefficient, naphthenic acid, oil pollutants

HIGHLIGHTS

- Investigating the behavior of the diffusion of naphthenic acid in water.
- Determination of the diffusion coefficient of naphthenic acid in water.
- An experimental method designed to determine the diffusion coefficient.

GRAPHICAL ABSTRACT



1. INTRODUCTION

Naphthenic acids are a group of polar organic carboxylic acids naturally existing in crude oil, and they are primarily cycloaliphatic carboxylic acids of 10–16 carbons (McKay *et al.* 1975; Qian *et al.* 2001; Clemente *et al.* 2003; Brown & Ulrich 2015). Naphthenic acids are the most significant environmental pollutants resulting from the extraction of crude oil from tar sand reserves. Research has shown that toxicity is higher in fish, while it is lower for invertebrates and algae. Additionally, the high concentration of these acids in crude oil causes corrosion in oil transportation pipelines and exploration and refining equipment, resulting in a reduced lifespan of these facilities. Naphthenic acids, which are amber-yellow and have a molecular weight of 170, have a solubility of 50 mg/L of water. Analyzing naphthenic acids in crude oils is challenging. Electrospray ionization mass spectrometry (ESI-MS) offers a rapid and efficient method for this purpose due to its low information requirements (Lochte & Littmann 1955; Havre 2002; Rogers *et al.* 2002; Oliveira *et al.* 2004; Campos *et al.* 2005; Clemente & Fedorak 2005; Woo *et al.* 2009; Wang & Kasperski 2010; Damasceno *et al.* 2014; Kinley 2015; Shah *et al.* 2015; Swigert *et al.* 2015; Adams *et al.* 2019).

Sitaraman *et al.* (1963) found in a study that the Wick–Chang equation was not accurate in systems in which water is a substance dissolved in different organic solvents. Albery *et al.* (1967) developed a simple and rapid method for measuring diffusion coefficients using a Stokes cell and a pH stat and it was shown that the diffusion coefficients of different acids were mainly a function of their molecular weight. Schramke *et al.* (1999) in research investigated the water diffusion coefficients for organic compounds at 25 °C on 108 organic compounds, and the water diffusion coefficients predicted using molecular connectivity indices were statistically compared to the values calculated using the molar volume estimation and alkyl chain length methods (Othmer & Thakar 1953; Wilke & Chang 1955; Hayduk & Laudie 1974).

Miyabea & Isogai (2011) in a study experimentally led to the presentation of an equation developed from the Wick–Chang equation which had more favorable results than the previous equations. Bidstrup & Geankoplis (1963) in an experiment using the diaphragm cell method and at a temperature of 25 °C measured the diffusion coefficients for several groups of carboxylic acids. In a study, Gharagheizi (2012) presented a three-parameter equation to calculate the diffusion coefficient of non-electrolyte compounds in water. In an experimental study, Manouchehrian Fard & Beiki (2016) measured the molecular diffusion coefficient of benzoic acid in water-based nanofluids containing gamma alumina nanoparticles. Hamada & de Anna (2021) proposed a new method for direct measurement of the diffusion coefficient of solutes in liquids. Their method was based on direct measurement of the spatial profile of the concentration of a considered tracer by optical techniques in a single emission, and the diffusion coefficient can be estimated based on the Einstein–Stokes relationship.

Since naphthenic acid is the most common cause of corrosion in refinery equipment and is classified as the most important environmental pollutant during oil extraction, it has been the focus of researchers in the past few years (Sotoudeh *et al.* 2023).

In this research, various methods have been examined, and the best method for determining the diffusion coefficient of naphthenic acid in water has been presented experimentally and in the laboratory. In past studies, there were some limitations in estimating the diffusion coefficient of naphthenic acid using mathematical equations, and no precise experimental or laboratory method was used to determine the diffusion coefficient of this substance. For a better understanding and the high importance of this substance, precise information such as the diffusion coefficient of this substance in water for environmental modeling and examining its exact behavior in nature is needed. One of the major challenges of this study is finding a suitable indicator to determine different concentrations of naphthenic acid in water.

2. MATERIALS AND METHODS

2.1. Materials

Commercial naphthenic acid technical grade (purity of 90–100%) was purchased from Sigma-Aldrich (with technical specifications in Table 1). For ultra-low range chemical oxygen demand (COD) determination by the Reactor Digestion Method. Hach Method 8000. Range: 0.7–40.0 mg/L COD. Pack of 150 vials. Distilled water was used in this experiment. No further purification of the chemicals was performed once they were received.

2.2. Equipment and setup

Molecular diffusion, as the name suggests, occurs slowly due to the collision of molecules, but if a magnet is placed inside this container for stirring, the diffusion of molecules will be done much faster. Armfield company's device, known as the liquid diffusion coefficient apparatus, calculates the diffusion coefficient based on cyclic diffusion and concentration changes by changing the numbers on the display of the electrical conductivity meter. However, since the solubility of naphthenic acid in water is very weak and the changes in the numbers of the electrical conductivity meter are very small, it is necessary to build a new device and method with the ability to identify heavy oil pollutants with poor solubility. In this research, first, by making solutions with different concentrations of naphthenic acid, the calibration chart of COD over concentration was obtained, and then the COD over time graph was obtained by experimenting; as a result, with the help of these two graphs, the diagram of concentration over time was obtained.

To observe the changes in the concentration of naphthenic acid in water, a new device was made, and the following methods were used to measure the concentration changes. This device consists of a bottle with a volume of 1 L of distilled water at a temperature of 25 °C. According to Figure 1, the desired tubular funnel was made by the laboratory equipment manufacturer. The specifications of the pipe were 15 cm long from the inlet side and 7 cm long from the outlet side. The specifications of the funnel on the outlet opening of the pipe were 3 cm high and 2.5 cm in diameter. The ratio of the diameter of the surface of the funnel for the exit of the substance should be 1/4 of the surface of a 1-L beaker (10 cm). After examining the different perforated sheets and the rate of naphthenic acid passing through them, the best dimensions that spread the diffusion slowly and normally in the water were the dimensions of 0.1 for the holes.

The volume of naphthenic acid poured into the U-shaped diffusion tube should be such that the liquid height on both sides of the tube is the same and equal to the level of the edge of the outlet funnel (Figure 2). To ensure proper diffusion, the outlet

Table 1 | Technical and laboratory specifications of the applied naphthenic acid (Sigma-Aldrich)

Parameter	General characteristic
Code	70,340
Molecular formula	C ₁₀ H ₁₈ O ₂
Quality	Level 100
Grade	Technical
Refractive index	n ₂₀ /D 1.45
Acid value	~230
Density	0.92 g/mL at 20 °C (lit.)

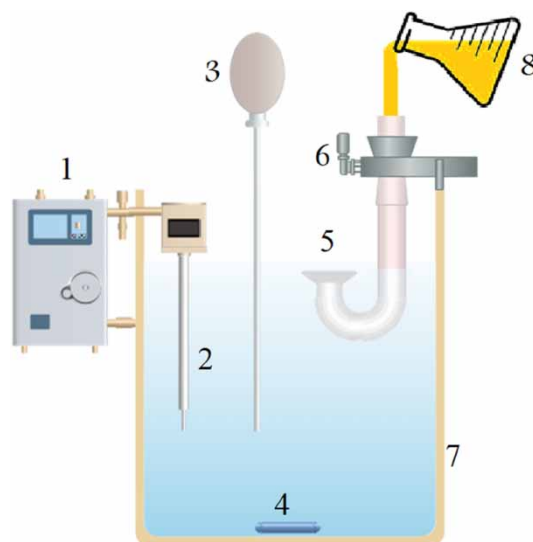


Figure 1 | Diffusion of naphthenic acid in water (1. pH meter, 2. pH meter electrode, 3. pipette for sampling, 4. magnet, 5. release funnel, 6. funnel holder, 7. distilled water tank, 8. naphthenic acid).



Figure 2 | Experiment to determine the diffusion coefficient of naphthenic acid in water.

funnel should be 5 mm below the submerged water level so that diffusion occurs smoothly with the help of water molecules. Rotating the magnetic stirrer at a low speed improves the diffusion process for more efficient sampling.

2.3. Concentration measurements

After examining various methods in the laboratory, such as the electrical conductivity meter, pH meter, oxidation reduction potential (ORP), and the COD method to better observe the changes in the concentration of naphthenic acid in the water, the COD method showed the best concentration changes, and in this experiment, this method was used to determine the diffusion coefficient.

After several tests to find the time to stabilize the concentration, it was decided to take 10 samples from the device with a laboratory pipette every 30 min at intervals of 4 hours. Sampling points were fixed and in the middle of the beaker. The samples removed with a pipette were poured into special COD vials and analyzed with the help of a DRB200 digital reactor

block for test 'N' tube (TNT) plus: 9×13 mm vial wells, 2×20 mm vial wells, 115 volts alternating current (VAC), and a Hach Lange DR5000 UV/VIS spectrophotometer, and the read numbers after comparing with the calibration chart showed different concentrations of naphthenic acid.

2.4. Diffusion coefficient calculation

Fick's laws of diffusion were derived by Adolf Fick in 1855. These laws describe the speed of diffusion and can be utilized for solving problems related to the diffusion coefficient. Fick's first law is used for steady-state diffusion (Fick 1855):

$$J = -D \frac{\partial \phi}{\partial x} \quad (1)$$

In this regard, J is the diffusion flux in ($\text{mol}/\text{m}^2\text{s}$), D is the diffusion coefficient in (m^2/s), ϕ is the concentration in (mol/m^3), and x is the diffusion distance in meters. The negative sign means that the direction of diffusion is down the concentration gradient, that is, from high concentration to low concentration. Fick's second law is used for unstable environments (Fick 1855):

$$\frac{\partial \phi}{\partial t} = D \frac{\partial^2 \phi}{\partial x^2} \quad (2)$$

Usually, the following equation is used to calculate the diffusion coefficient of a substance in water, and its method of operation is based on an electrical conductivity meter (McCabe *et al.* 2005):

$$D = \frac{4VX}{N\pi d^2 MCm} \left(\frac{dk}{dt} \right) \quad (3)$$

In this regard, D is the diffusion coefficient in (cm^2/s), V is the volume of the cylinder (cm^3), X is the length of the diffusion layer (cm), d is the diameter of the mesh holes (cm), N is the number of holes in the mesh plate, M is the concentration of the solution in molar, dk/dt is the slope of the graph of change in electrical conductivity over time, and Cm is the conversion factor of the change in electrical conductivity to the change in solution concentration (slope of the COD/C diagram in molar).

3. RESULTS AND DISCUSSION

3.1. Diffusion coefficient determination

The total diffusion coefficient in hydraulics is shown by D and is defined as the sum of molecular diffusion and turbulent diffusion. Since the diffusion coefficient of naphthenic acid has a direct effect on the modeling of its diffusion and advection in water, therefore, this number is needed for modeling work and future studies, but since the diffusion coefficient has not been calculated for naphthenic acid so far, in this experiment, the diffusion coefficient of this substance in water was determined experimentally and in a laboratory. After examining different methods to observe the changes in the concentration of naphthenic acid in water, the COD method showed the best results, and this method was used to show the changes in the concentration of naphthenic acid in water.

3.2. Solution preparation

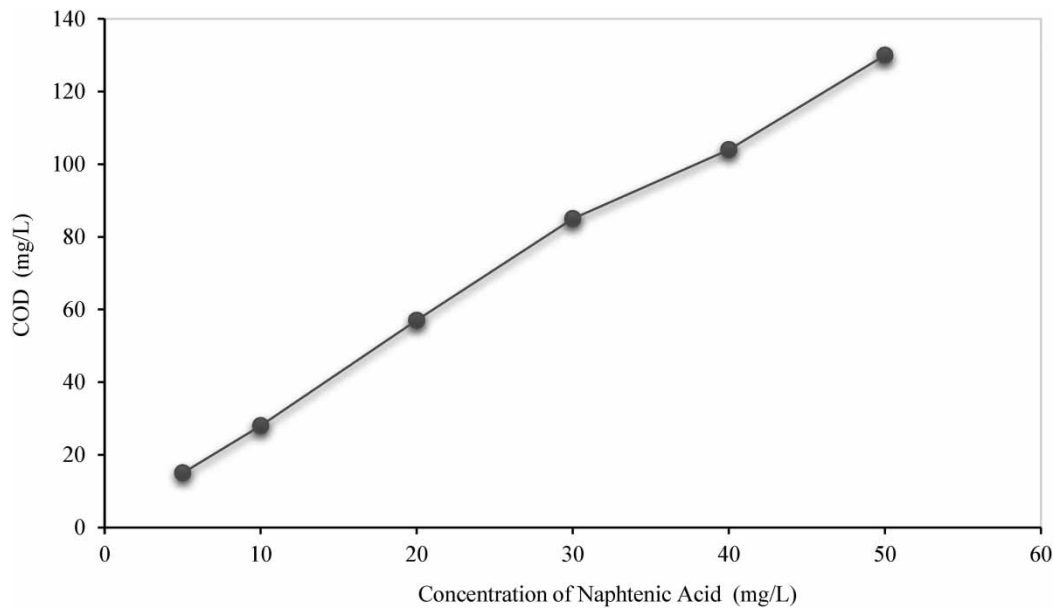
Since the solubility of naphthenic acid in water is 50 mg per liter, first 0.05 g of naphthenic acid is dissolved in 1 L of distilled water to make a solution of 50 mg/L, and then with the help of the obtained solution (Table 2), the solutions needed for calibration are made as discussed below.

3.3. Calibration and concentration measurement

For calibration, first, solutions with different concentrations of 5–50 mg/L of naphthenic acid were made, and 2 cc of each concentration was poured into special COD vials and then placed in the DRB200 reactor for 120 min. It was taken to reach a temperature of 150 °C and after cooling the vials, the samples were placed in the DR5000 spectrophotometer, and the COD number was read at specified times to obtain the COD of these concentrations (see Figure 3).

Table 2 | Different concentrations made from the naphthenic acid solution

Concentration made (mg/L)	Solution 50 mg/L (cc)	Water (cc)
5	10	90
10	20	80
20	40	60
30	60	40
40	80	20
50	100	0

**Figure 3** | COD calibration graph for different concentrations of naphthenic acid.

To prepare the standard solution, first, 2 cc of distilled water should be poured into a COD vial, and before placing the concentration vials in the DR5000 spectrophotometer, this standard vial should be placed in the device, and the ZERO option should be selected. The COD codes used in the device were 430 and 150 in the low range mode. Before placing the vials in the device, make sure they are clean and dry.

After conducting the experiment to determine the diffusion coefficient of naphthenic acid in water and completing the sampling at specified times, the COD values at different times were obtained according to Figure 4, and the COD over time graph was plotted. Additionally, the slope of this graph is the same as $dcod/dt$, which is used to solve Equation (4) for determining the diffusion coefficient.

To plot the concentration graph over different times, the obtained COD values at specified times must be placed into the equation of the calibration graph COD over concentration. This will yield the concentration over time as shown in Figure 5, indicating the gradual increase in the concentration of naphthenic acid in water over different times.

3.4. Diffusion coefficient calculation

Equation (3) is used to determine the laboratory diffusion coefficient as mentioned, but in this method, COD was used instead of the electrical conductivity meter to show the concentrations in different time intervals. Therefore, in Equation (3), dk should be replaced by $dCOD$ and rewritten as follows:

$$D = \frac{4VX}{N\pi d^2 MCm} \left(\frac{dCOD}{dt} \right) \quad (4)$$

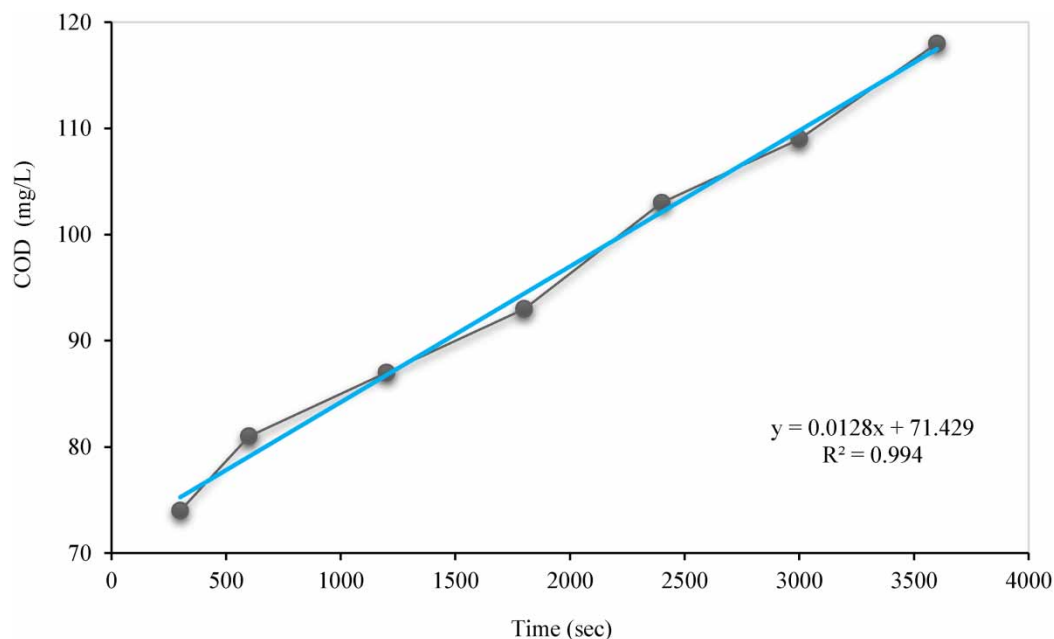


Figure 4 | COD of collected samples of naphthenic acid at different times.

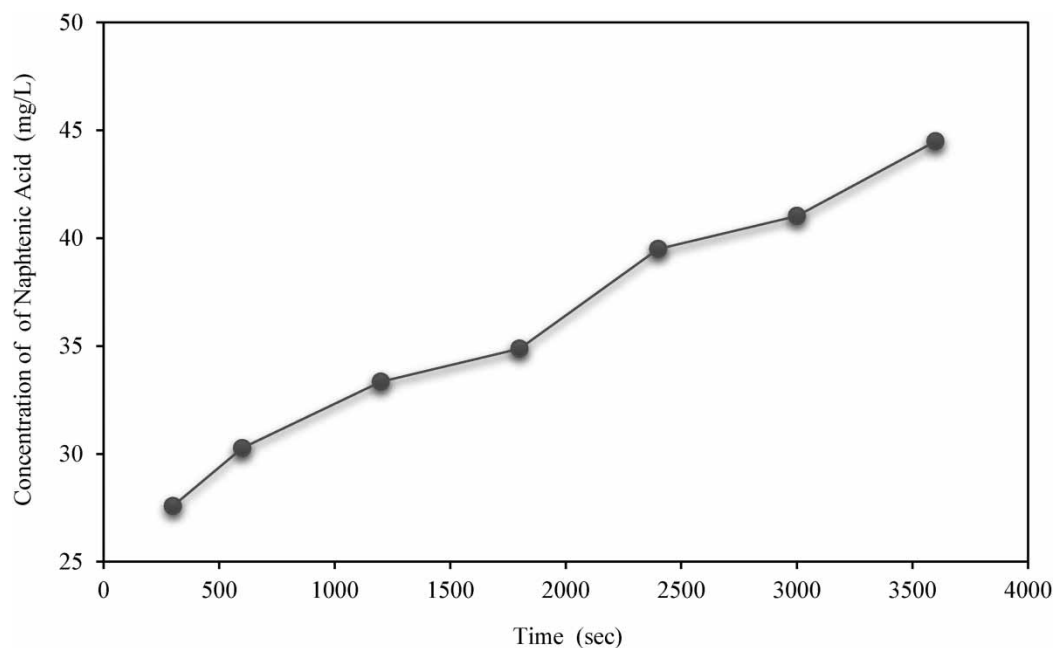


Figure 5 | Graph of naphthenic acid concentration changes in water at different times.

Equation (4) relates the rate of COD change over time to the concentration of naphthenic acid, allowing us to calculate the diffusion coefficient based on experimental data. As can be seen, all the variables in Equation (4) are similar to the variables in Equation (3), and only the two parameters C_m and dk/dt have changed. Therefore, in Equation (4), C_m is the slope of the graph of COD changes versus concentration in molar terms, and d_{COD}/dt is the slope of the graph of COD changes over time. Therefore, according to Figure 6, we need to plot the graph of COD values against concentration in molar terms. The slope of this graph, which is the number C_m in Equation (4), is needed to obtain the diffusion coefficient.

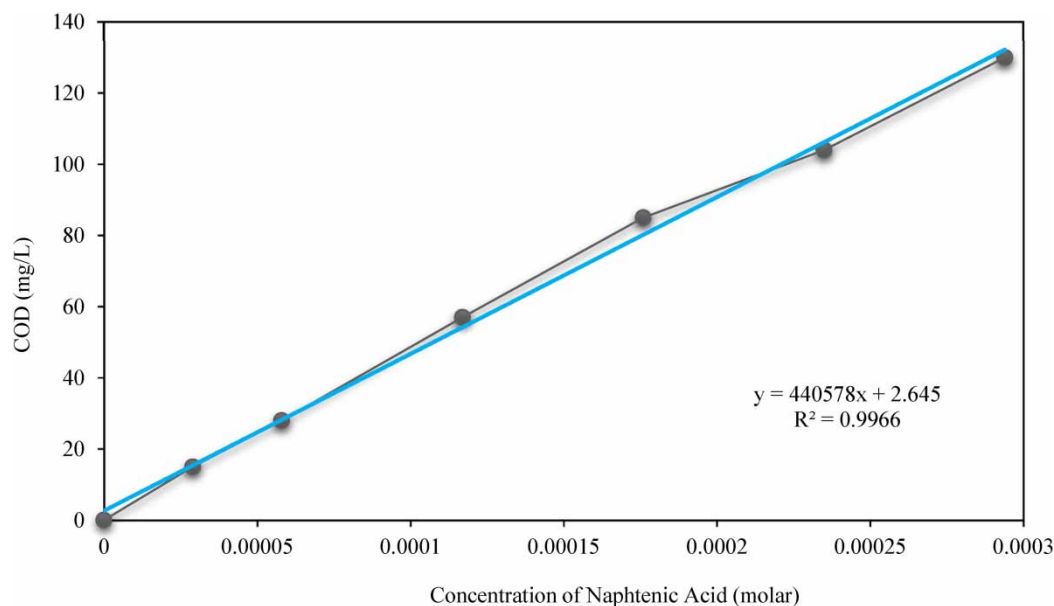


Figure 6 | COD calibration graph for different concentrations of naphthenic acid to molar.

As $Cm = 440,578$; $V = 1,000 \text{ m}^3$; $X = 0.5 \text{ cm}$; $d = 0.1 \text{ cm}$; $N = 500$; $\frac{d\text{COD}}{dt} = 0.0128$; and:

$$M = 10 \times \text{concentration percentage} \times \text{density/molecular weight} = \frac{10 \text{ pd}}{M} = 5.30 \quad (5)$$

By placing the obtained variables in Equation (4), the diffusion coefficient of naphthenic acid in water was obtained as $0.69 \times 10^{-9} \text{ m}^2/\text{s}$.

3.5. pH changes

Before the experiment, the pH of distilled water was measured, and it was equal to 7.1. Then, at the beginning of the experiment and the release of naphthenic acid, the pH value began to decrease so that at the end of the experiment, the pH value reached 5.3, and this indicates the increase in the release of naphthenic acid in water over time and its acidification (Figure 7).

Finally, according to the diffusion coefficients of similar acids such as citric acid $0.68 \times 10^{-9} \text{ m}^2/\text{s}$, succinic acid $0.93 \times 10^{-9} \text{ m}^2/\text{s}$, propionic acid $1.18 \times 10^{-9} \text{ m}^2/\text{s}$, and stearic acid $0.61 \times 10^{-9} \text{ m}^2/\text{s}$, it is possible to understand the accuracy of the diffusion coefficient of naphthenic acid in water. Of course, this small number means the solubility and poor diffusion of naphthenic acid in water (Liu *et al.* 2004; Xu & Yang 2019).

4. CONCLUSION

In this research, for the first time, the diffusion coefficient of naphthenic acid in water was obtained using experimental and laboratory methods to serve as a precursor for future studies on the modeling and behavior of this substance in the natural environment. The diffusion coefficient of naphthenic acid in distilled water at a temperature of 25°C was found to be $0.69 \times 10^{-9} \text{ m}^2/\text{s}$, indicating the weak penetration of this substance in water. The innovation of our research was the introduction of a new laboratory method for determining the diffusion coefficient of polar organic compounds that have poor solubility in water. This research also led to the development of the laboratory diffusion coefficient empirical equation in the Unit Operations textbook (McCabe *et al.* 2005). After examining the diffusion coefficients of substances similar to naphthenic acid, which have nearly similar properties and carbon rings, one can ascertain the accuracy of the diffusion coefficient of naphthenic acid in water. It is worth mentioning that in this research, three other methods, such as ORP, pH meter, and electrical conductivity meter, were used, but the chemical oxygen demand (COD) method provided the best range of concentration changes over different times within an appropriate numerical range.

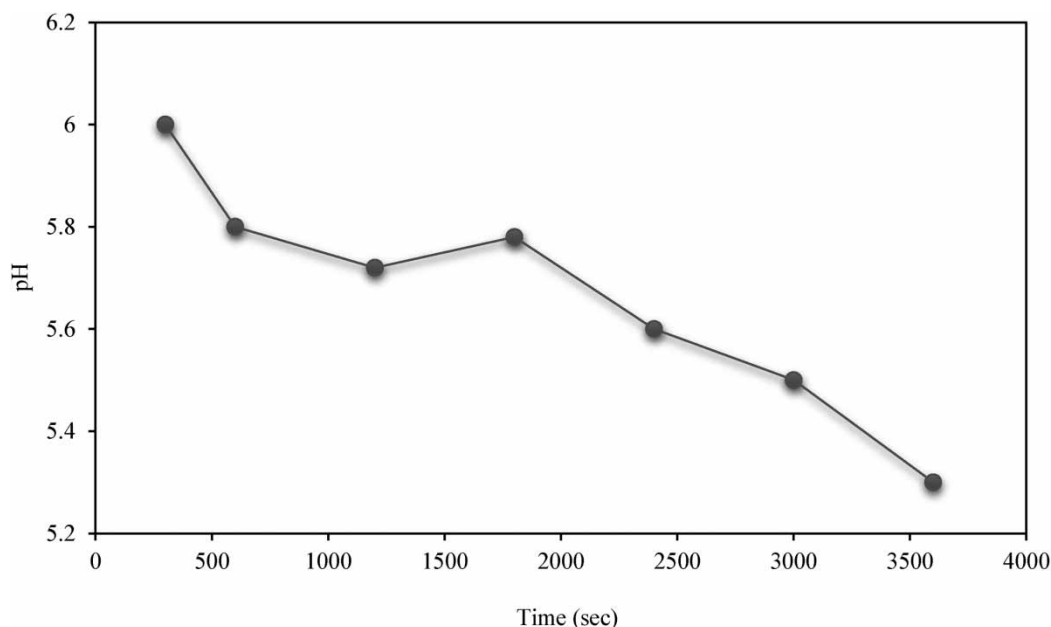


Figure 7 | Graph of pH changes in water during the experiment to determine the diffusion coefficient of naphthenic acid.

Since naphthenic acid is one of the most important oil pollutants during the crude oil extraction phase, a large amount of this substance enters the environment and seas along with the water that is extracted from the well during crude oil extraction. Therefore, for environmental research such as modeling its advection, behavior, and fate in the natural environment, precise information like the diffusion coefficient of this substance in water is required. However, in the past, due to a lack of understanding of the importance of the subject, researchers only used mathematical equations to estimate the diffusion coefficient of this substance. For future studies, it is suggested to work on modeling the diffusion and advection of naphthenic acid in aquatic environments and seas.

AUTHORS CONTRIBUTIONS

Y.S. conceptualized and validated the project, performed in experiment, investigated the process, contributed to software, wrote the original draft, visualized the work, developed the methodology, reviewed the article; M.H.N. supervised the work, developed the methodology, conceptualized the project, investigated the process, wrote and reviewed and edited the article; A.K. supervised the work, developed the methodology, conceptualized the project, wrote and reviewed and edited the article; M.B. developed the methodology and performed in experiment.

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DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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